



MCT5210 – Control Engineering III

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Modern Control

- Due to requirements of complex tasks and accuracy, modern engineering systems have increased complexity.
- Complex systems may have **multiple inputs** and **multiple outputs** and may also be **time varying**.
- Modern control theory has been developed to cater for increased complexity and performance requirements as well as employ the use of large scale computers which are now easily accessible.
- Modern control theory is based on the concept of **state**.
- The **state** of a dynamic system is the **smallest set of variables** (called state variables) such that knowledge of these variables at $t = t_0$, together with knowledge of the input for $t \geq t_0$, completely determines the behavior of the system for any time $t \geq t_0$.
- Modern control theory is applicable to **MIMO** systems, which may be **linear** or **nonlinear**, time invariant or time varying, unlike conventional control which is applicable only to LTI SISO systems.



State Vectors and State Space

State Vector: A state vector is a vector that uniquely determines the system state $\mathbf{x}(t)$ for any time $t \geq t_0$, once the state at $t = t_0$ is given and the input $u(t)$ for $t \geq t_0$ is specified.

That is, if n state variables are needed to completely describe the behavior of a given system, then these n state variables can be considered as the n components of a state vector $\mathbf{x}$.

State Space: The n -dimensional space whose coordinate axes consist of the $x_1, x_2, \dots, x_n$ axis, where $x_1, x_2, \dots, x_n$ are state variables, is called a **state space**. Any state can be represented by a point in the state space.



State-Space Equations

- State-space analysis is concerned with variables that are involved in the **modeling** of dynamic systems.
- These are **input** variables, **output** variables, and **state** variables.
- The dynamic system must involve elements that **memorize** the values of the input for $t \geq t_0$.
- The **outputs** of **integrators** can be considered as the variables that define the **internal state** of a dynamic system, since integrators serve as **memory** devices in a continuous-time control system .
- Thus the outputs of integrators serve as state variables.
- The **number** of **state** variables to completely define the dynamics of a system is equal to the **number** of **integrators** involved in the system.



State-Space Equations

- Consider a **MIMO** system involving n **integrators** with r **inputs** $u_1, u_2, \dots, u_r$ and m **outputs** $y_1, y_2, \dots, y_m$. By defining the n outputs of the integrators as state variables $x_1, x_2, \dots, x_n$, then the **system** may be described by

$$\begin{aligned}\dot{x}_1 &= f_1(x_1, x_2, \dots, x_n; u_1, u_2, \dots, u_r) \\ \dot{x}_2 &= f_2(x_1, x_2, \dots, x_n; u_1, u_2, \dots, u_r) \\ &\vdots \\ \dot{x}_n &= f_n(x_1, x_2, \dots, x_n; u_1, u_2, \dots, u_r)\end{aligned}\tag{1}$$

And the **outputs** given by

$$\begin{aligned}y_1 &= g_1(x_1, x_2, \dots, x_n; u_1, u_2, \dots, u_r) \\ y_2 &= g_2(x_1, x_2, \dots, x_n; u_1, u_2, \dots, u_r) \\ &\vdots \\ y_m &= g_m(x_1, x_2, \dots, x_n; u_1, u_2, \dots, u_r)\end{aligned}\tag{2}$$



State-Space Equations

- If we define

$$\mathbf{x}(t) = \begin{bmatrix} x_1(t) \\ x_2(t) \\ \vdots \\ x_n(t) \end{bmatrix}, \quad \mathbf{f}(\mathbf{x}, \mathbf{u}, t) = \begin{bmatrix} f_1(x_1, x_2, \dots, x_n; u_1, u_2, \dots, u_r; t) \\ f_2(x_1, x_2, \dots, x_n; u_1, u_2, \dots, u_r; t) \\ \vdots \\ f_n(x_1, x_2, \dots, x_n; u_1, u_2, \dots, u_r; t) \end{bmatrix},$$
$$\mathbf{y}(t) = \begin{bmatrix} y_1(t) \\ y_2(t) \\ \vdots \\ y_m(t) \end{bmatrix}, \quad \mathbf{g}(\mathbf{x}, \mathbf{u}, t) = \begin{bmatrix} g_1(x_1, x_2, \dots, x_n; u_1, u_2, \dots, u_r; t) \\ g_2(x_1, x_2, \dots, x_n; u_1, u_2, \dots, u_r; t) \\ \vdots \\ g_m(x_1, x_2, \dots, x_n; u_1, u_2, \dots, u_r; t) \end{bmatrix}, \quad \mathbf{u}(t) = \begin{bmatrix} u_1(t) \\ u_2(t) \\ \vdots \\ u_r(t) \end{bmatrix}$$

Then Equations (1) and (2) become

$$\dot{\mathbf{x}}(t) = \mathbf{f}(\mathbf{x}, \mathbf{u}, t) \quad (3)$$

$$\mathbf{y}(t) = \mathbf{g}(\mathbf{x}, \mathbf{u}, t) \quad (4)$$



State-Space Equations

- If vector functions $\mathbf{f}$ and/or $\mathbf{g}$ involve time t explicitly, then the system is called a **time varying**.
- Equations (3) and (4) can be linearised about the point therefore leading to the following linearised state space and output equations

$$\begin{aligned}\dot{\mathbf{x}}(t) &= \mathbf{A}(t)\mathbf{x}(t) + \mathbf{B}(t)u(t) \\ \mathbf{y}(t) &= \mathbf{C}(t)\mathbf{x}(t) + \mathbf{D}(t)u(t)\end{aligned}$$

where:

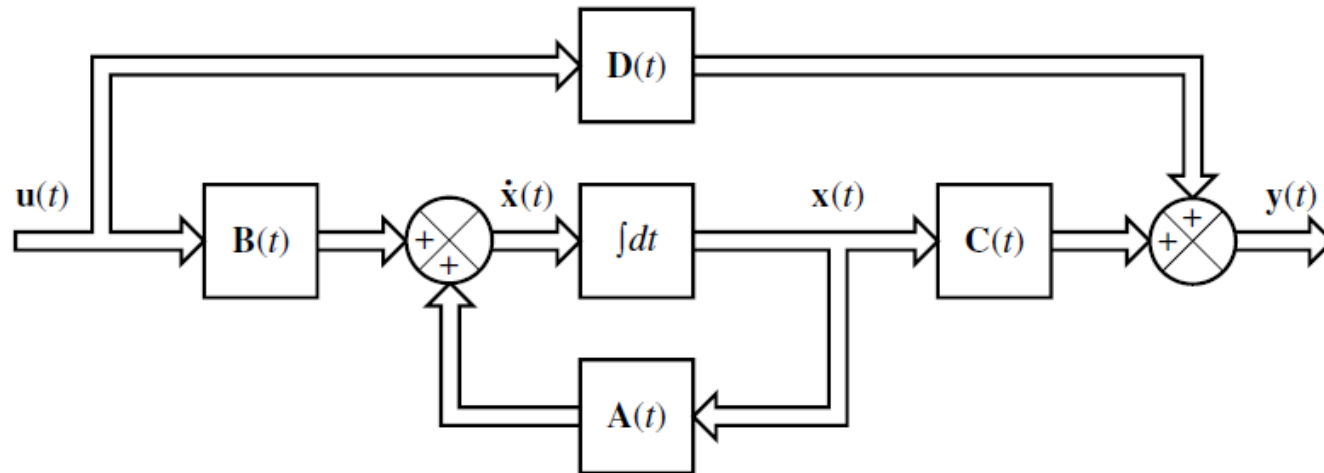
$\mathbf{A}(t)$ is referred to as the state matrix $\mathbf{B}(t)$ the input matrix, $\mathbf{C}(t)$ the output matrix, and $\mathbf{D}(t)$ the direct transmission matrix.

- If vector functions $\mathbf{f}$ and $\mathbf{g}$ do not involve time t explicitly then the system is called a **time-invariant** system.

$$\begin{aligned}\dot{\mathbf{x}}(t) &= \mathbf{A}\mathbf{x}(t) + \mathbf{B}u(t) \\ \mathbf{y}(t) &= \mathbf{C}\mathbf{x}(t) + \mathbf{D}u(t)\end{aligned}$$



State-Space Representation



State space block diagram representation of a linear, continuous time control system



Example 1: Mass, Spring and Damper

Consider the SISO mechanical system shown below. The external force $u(t)$ is the input to the system, and the displacement $y(t)$ of the mass is the output.

- The system equation can be expressed as
$$m\ddot{y} + b\dot{y} + ky = u$$
- The system is of second order and can thus be modelled with **two integrators**.
- Defining state variables as

$$x_1(t) = y(t)$$

$$x_2(t) = \dot{y}(t)$$

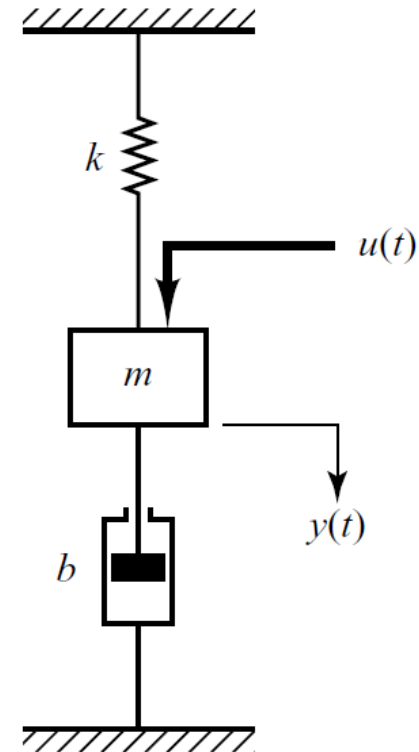
Therefore

$$\dot{x}_1 = x_2$$

$$\dot{x}_2 = -\frac{k}{m}x_1 - \frac{b}{m}x_2 + \frac{1}{m}u$$

and

$$y = x_1 \quad (\text{output equation})$$





Example 1: Mass, Spring and Damper

The state equation for the system can be expressed in matrix form as

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -\frac{k}{m} & -\frac{b}{m} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} 0 \\ \frac{1}{m} \end{bmatrix} u$$

And the output equation as

$$y = \begin{bmatrix} 1 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

Which is in the form

$$\dot{x} = \mathbf{A}x + \mathbf{B}u$$

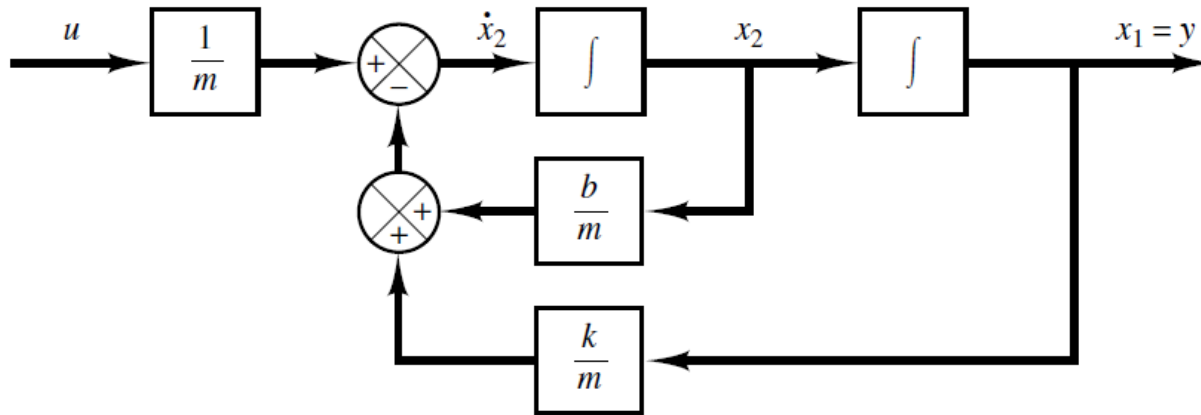
$$y = \mathbf{C}x + \mathbf{D}u$$

Where

$$\mathbf{A} = \begin{bmatrix} 0 & 1 \\ -\frac{k}{m} & -\frac{b}{m} \end{bmatrix}, \mathbf{B} = \begin{bmatrix} 0 \\ \frac{1}{m} \end{bmatrix}, \mathbf{C} = \begin{bmatrix} 1 & 0 \end{bmatrix}, \mathbf{D} = 0$$



Example 1: Block Diagram





Example 2: Simple Pendulum

Consider the simple pendulum shown in the figure.

- The motion of the system occurs in the vertical plane.
- From Newton's second law
$$F = ma = -mg \sin \theta \Rightarrow a = -g \sin \theta$$
- The linear acceleration along the arc is related to the motion along the arc with length s

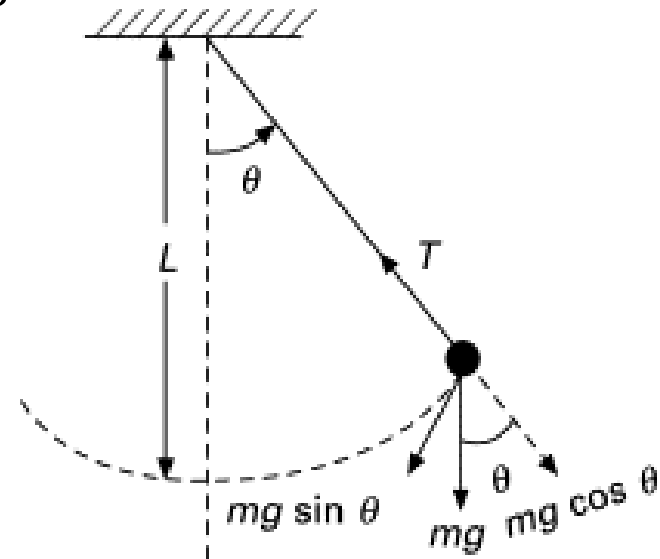
$$s = L\theta, a = \frac{d^2 s}{dt^2} = L \frac{d^2 \theta}{dt^2}$$

Thus

$$L \frac{d^2 \theta}{dt^2} = -g \sin \theta$$

Therefore

$$\frac{d^2 \theta}{dt^2} + \frac{g}{L} \sin \theta = 0$$



Let

$$x_1 = \theta$$

$$x_2 = \dot{\theta}$$

Thus

$$\dot{x}_1 = x_2$$

$$\dot{x}_2 = +\frac{g}{L} \sin x_1$$